STRESS Dataset Report

1.0 *STRESS* Variable Assessment

The *STRESS* variable contains the number of stressful events an adolescent reporting in the past year across the United States. A comprehensive assessment – descriptive statistics, a histogram, and Q-Q plot – of the response variable (*STRESS*) was performed to identify the distribution. The table below summarizes the key descriptive statistics for the *STRESS* variable (n = 651). The mean stress score is 1.73, while the median is 1 and the mode is 0. The considerable gap between the mean and mode, combined with a range of 0 to 9, suggests a distribution that is skewed to the right rather than symmetric.

n	Mean	Median	Mode	Variance	Std Dev	Min	Max	Range
651	1.73	1	0	3.42	1.85	0	9	0–9

The histogram below plots the frequency of each *STRESS* score from 0 to 9. The distribution shows a clear and pronounced right (positive) skew, with the tallest bars concentrated at the lower end of the scale. There are several notable observations in the histogram. First, 221 respondents (34%) reported a *STRESS* score of 0, indicating the large group. Second,128 (20%) scored 1 and 125 (19%) scored 2, meaning 73% of respondents scored 2 or below. Third, frequencies decline steadily as scores increase, with only 27 respondents (4%) scoring 6 or higher. Finally, the long right tail indicates a small proportion of individuals experiencing considerably higher stress, indicating the distribution is not normal.

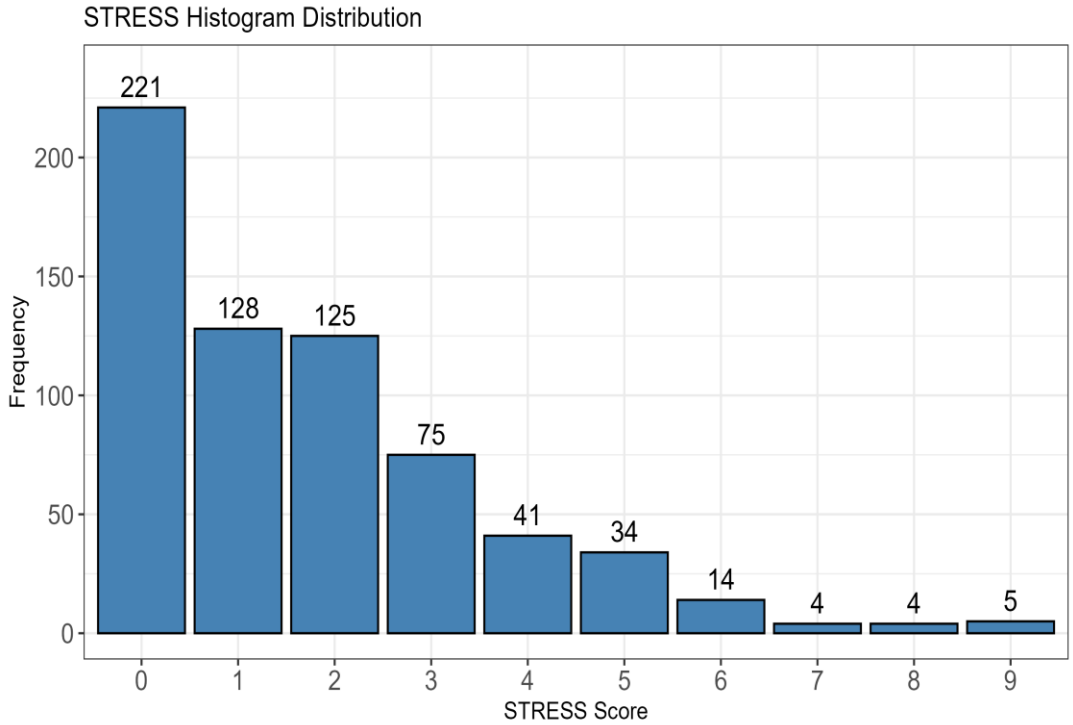

Figure 1. Distribution of STRESS variable (n = 651).

A Normal Q-Q (Quantile-Quantile) plot compares the sample distribution against what would be expected if the data were perfectly normally distributed. Points falling along the diagonal reference line indicate normality; systematic departures from the line indicate non-normality. The Q-Q plot below reveals two clear departures from normality. First, large mass of points lies flat along the bottom-left of the plot, far below the reference line. This reflects the 221 respondents who scored zero, a prominent feature that a continuous normal distribution cannot accommodate. Second, the points curve above the reference line, indicating that the highest stress scores are more extreme than a normal distribution would predict.

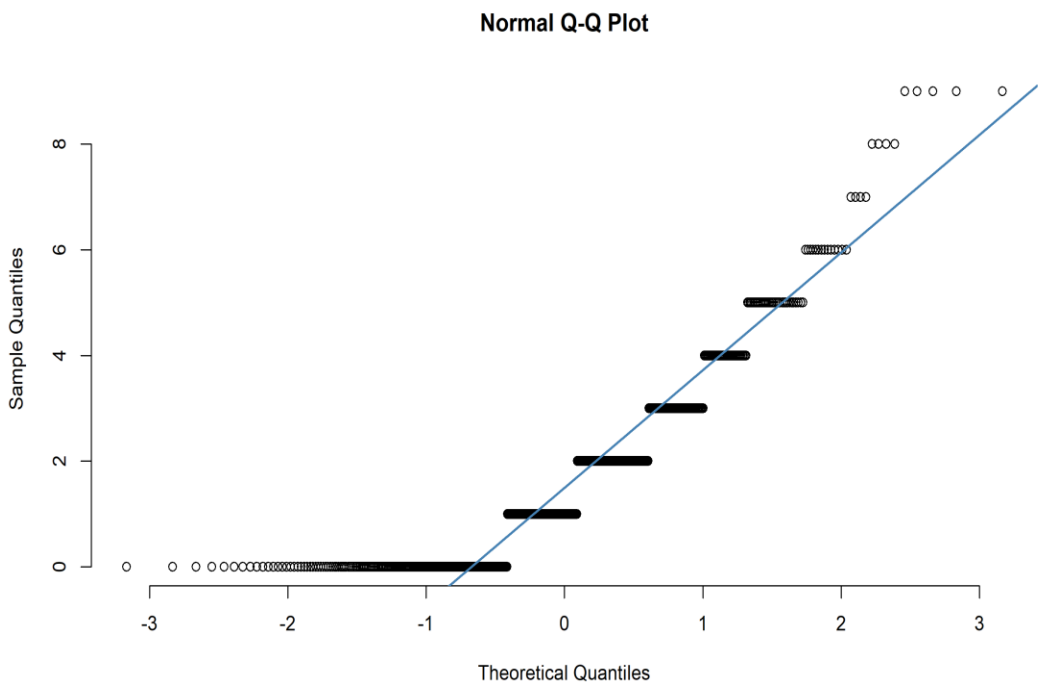

Figure 2. Q-Q Plot of STRESS variable (n = 651).

2.0 OLS Regression

2.1 Coefficient Interpretation

An Ordinary Least Squares (OLS) regression model was fitted to predict *STRESS* scores (range 0–9) using four explanatory variables: family cohesion (*COHES*), self-esteem (*ESTEEM*), academic grades (*GRADES*), and school attachment (*SATTACH*). The table below presents the estimated regression coefficients, standard errors, t-values, and significance levels.

Term	Estimate	Std. Error	t-value	p-value
(Intercept)	5.71281	0.58118	9.830	< 0.001
COHES	−0.02319	0.00703	−3.298	0.00103
ESTEEM	−0.04129	0.01933	−2.136	0.03305
GRADES	−0.04170	0.02352	−1.773	0.07670
SATTACH	−0.03042	0.01412	−2.154	0.03160

All four predictors are negatively associated with *STRESS*, consistent with the expectation that higher cohesion, self-esteem, grades, and school attachment are protective against stress. Three of the four predictors (*COHES*, *ESTEEM*, *SATTACH*) reach conventional significance at $\alpha = 0.05$. *GRADES* approaches but does not meet the threshold ($p = 0.077$). The intercept is large (5.71) and highly significant, representing the expected stress score when all predictors are zero, a value that lies well outside the observed predictor range and should not be interpreted directly.

2.2 Overall Model Fit

Despite reaching statistical significance ($F_{(4, 646)} = 14.65$, $p < 0.001$), the model explains only 8.3% of the variance in *STRESS* scores ($R^2 = 0.0832$). This is a very weak effect and indicates that the four predictors, whilst statistically relevant, account for only a small fraction of the variation in stress across respondents. The residual standard error of 1.776 means the average prediction error is nearly two points on a nine-point scale, practically substantial for decision-making purposes.

2.3 Predicted Values Histogram

A histogram of the predicted values ($\hat{Y}$) from OLS regression model was performed to identify the distribution.

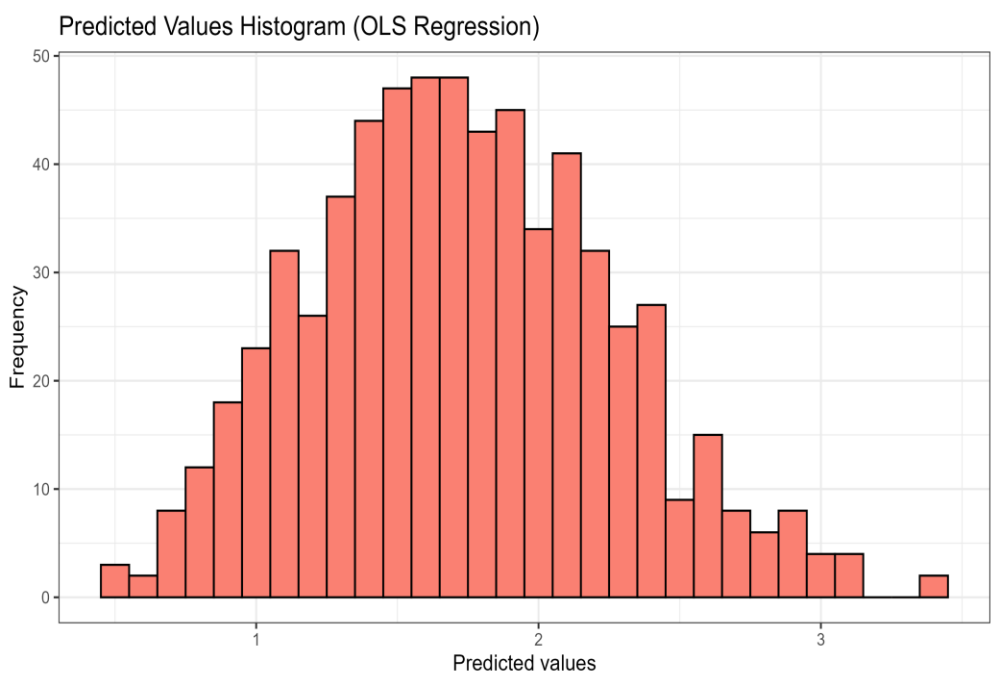

Figure 3. Distribution of OLS predicted values for STRESS (n = 651).

The predicted values range from approximately 0.5 to 3.3 and follow a roughly bell-shaped, symmetric distribution. This creates an immediate and fundamental problem: the observed *STRESS* data are right-skewed integers concentrated heavily at 0, 1, and 2, yet the model produces a continuous, near-normal spread of predictions. The model structurally misrepresents the outcome variable in two critical ways. First, OLS can generate negative predicted values (or

values well below zero) for respondents at the lower end of the predictor range, which is impossible for a count variable. Second, the smooth, symmetric predicted distribution bears no resemblance to the heavily right-skewed, discrete observed distribution, meaning the model's predictions are systematically misleading.

2.4 Model Diagnostic Assessment

Standard diagnostic plots in were created to assess assumptions of the OLS-fitted model, including Pearsons residual for each individual predictor, residuals versus fitted values, Cook's distance, Hat-values (leverage), and Bonferroni outlier tests.

For a well-specified OLS model, residuals should be randomly scattered around zero with no pattern. The plots against each individual predictor show the spread of residuals is clearly not constant as the vertical range of residuals is far wider in the middle and upper ranges of each predictor, suggesting heteroscedasticity. The residuals versus fitted values plot reveals a pronounced fan shape: residual variance is narrow at low fitted values and expands substantially at higher fitted values. This is a textbook signature of heteroscedasticity. Additionally, Non-Constant Variance Score test was performed and heavily significant ($\chi^2(1) = 46.34$, $p < 0.001$), providing more evidence of heteroscedasticity.

Cook's distance measures the overall influence of each observation on the fitted model. Cases flagged above the conventional threshold ($D > 4/n \approx 0.006$) are considered influential. Case 134 was clearly the single most influential data point, and Case 242 had high influence due to a combination of moderate-to-high leverage and a large positive residual (studentized residual ≈ 3.5). Hat-values quantify how far an observation's predictor values are from the centroid of the predictor space. Cases 569 and 632 show the highest hat-values (≈ 0.038–0.040), placing them well outside the predictor cloud. Although their residuals are not extreme, their high leverage means they can silently distort coefficient estimates.

The Bonferroni-corrected p-values panel confirms that cases 134 and 411 have statistically significant studentized residuals after adjustment for multiple comparisons, classifying them as formal outliers in the context of this model.

3.0 Transformed OLS Regression

3.1 Coefficient Interpretation

A natural log transformation was applied to the *STRESS* outcome — specifically LN(*STRESS* + 1), using log1p() to handle the large number of zero values — before fitting an OLS regression with the same four predictors (*COHES*, *ESTEEM*, *GRADES*, *SATTACH*). The rationale for this transformation is to reduce right skew and stabilize residual variance. The +1 offset is necessary because ln(0) is undefined, and 221 respondents (34%) have STRESS = 0. This maps zeros to ln(1) = 0 on the transformed scale, preserving them while avoiding undefined values. However, this creates its own issue: all zero-stress respondents collapse to a single point (LN(Y) = 0), which does not genuinely resolve the excess-zero problem. The table below reports the estimated regression coefficients, standard errors, t-values, and significant levels.

Term	Estimate	Std. Error	t-value	p-value
(Intercept)	2.07332	0.20910	9.916	< 0.001
COHES	−0.00795	0.00253	−3.142	0.00175
ESTEEM	−0.01092	0.00696	−1.569	0.11706
GRADES	−0.01434	0.00846	−1.694	0.09072
SATTACH	−0.01128	0.00508	−2.220	0.02674

All coefficients remain negative, consistent with Model 1. However, the significance levels of the predictors have changed. *COHES* and *SATTACH* remain significant ($p = 0.002$), with *COHES* as the most robust predictor. *ESTEEM* is no longer statistically significant ($p = 0.117$), having been significant in Model 1 ($p = 0.033$). The transformation has diluted its signal. *GRADES* continues to approach but does not reach significance ($p = 0.091$). Additionally, Coefficient interpretation is now more complex: a one-unit increase in each predictor is associated with a multiplicative change in (*STRESS* + 1) on the original scale, not an additive change in *STRESS* itself. For example, a one-unit increase in *COHES* is associated with a 0.79% decrease in (*STRESS* + 1), all else equal. This is less intuitive for management reporting.

3.2 Model Fit

The R^2 of Model 2 (7.15%) is slightly lower than Model 1 (8.32%), meaning the transformation did not improve explanatory power, it reduced it. This is expected when a transformation misaligns with the true data-generating process. Both models explain less than 9% of variance in stress, confirming weak overall fit regardless of transformation choice.

3.3 Predicted Values Histogram

A histogram of the predicted values ($\hat{Y}$) from natural log transformed. OLS regression model was performed to identify the distribution.

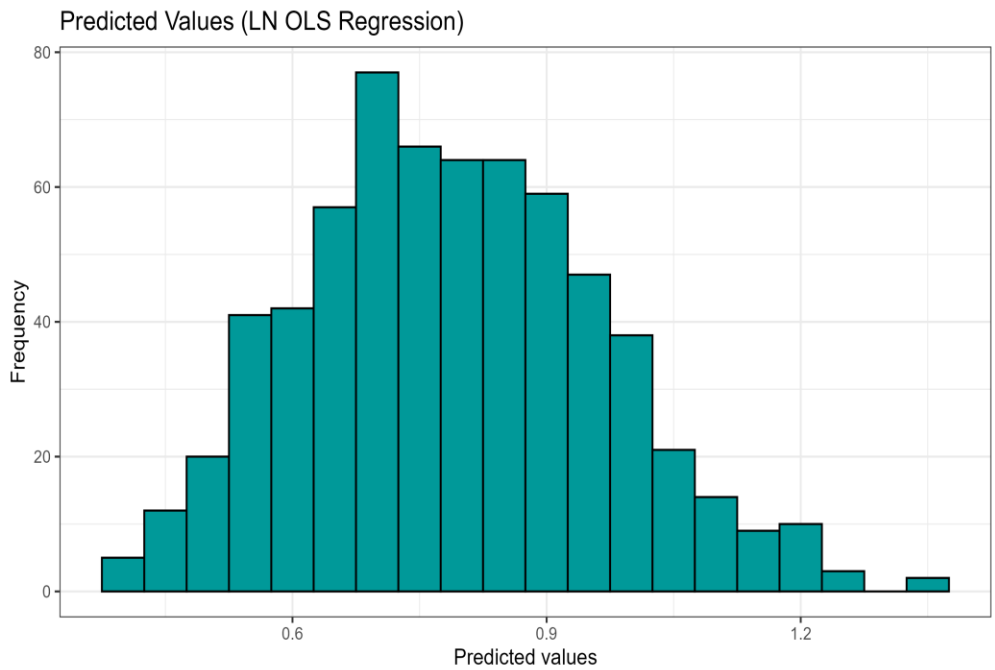

Figure 4. Distribution of LN OLS predicted values for LN(STRESS+1) (n = 651).

The predicted values on the log scale range from approximately 0.45 to 1.25 and follow a roughly bell-shaped, slightly right-skewed distribution, more symmetric than the original Model 1 predictions but still a poor reflection of the actual outcome distribution. Several key issues remain. First, the predictions are on the log scale (LN units), not the original *STRESS* scale. To interpret them meaningfully, they must be back-transformed via $\exp(\hat{Y}) - 1$. Second, back-transformation introduces a systematic bias: $\exp(E[LN(Y)]) \neq E[Y]$. The expected value on the original scale is not simply $\exp(\hat{Y})$. A smearing correction or alternative estimator would be required.

3.4 Diagnostic Assessment

Compared to Model 1, the extreme outliers (residuals of +6 or more) have disappeared, indicating a genuine improvement. The residual range is now approximately −1.2 to +1.4, substantially compressed. However, the residuals vs. each predictor still show unequal spread, with greater variance in the middle and upper ranges. Additionally, the residuals vs. the predicted values demonstrate a pronounced triangular or wedge pattern, rather than a fan-shape seen in the OLS model. This is seen as a dense horizontal band of negative residuals sits near −1 across all fitted values (reflecting the floor created by the zero-stress respondents mapped to LN = 0), while the upper cloud of positive residuals tapers from left to right. This is a different form of heteroscedasticity, not an absence of it.

The log transformation has meaningfully reduced the influence of extreme cases as the maximum Cook's Distance dropped from 0.0807 (OLS) to 0.0268 (LN OLS). Additionally, no cases register as statistically significant outliers under the Bonferroni correction (all p-values ≈

1.0), compared to two formal outliers (cases 134 and 411) in the OLS regression. Though, Case 134 remains the most influential point and its leverage has been substantially tempered.

4.0 Poisson Regression

4.1 Coefficient Interpretation

A standard Poisson count model was fitted to predict *STRESS* scores with the same predictors as the OLS models. The table below highlights the estimate, standard error, z-score, p-value, and the incident rate ratio (IRR) of coefficient in the fitted model.

Term	Estimate	Std. Error	z-value	p-value	Interpretation (IRR)
(Intercept)	2.7345	0.2341	11.683	< 0.001	exp(2.7345) = 15.40; baseline rate when all X = 0
COHES	−0.01292	0.00289	−4.466	< 0.001	Each +1 unit: STRESS rate × exp(−0.01292) = × 0.9872 (↓ 1.28%)
ESTEEM	−0.02369	0.00804	−2.947	0.00321	Each +1 unit: STRESS rate × 0.9765 (↓ 2.35%)
GRADES	−0.02347	0.00987	−2.379	0.01735	Each +1 unit: STRESS rate × 0.9768 (↓ 2.32%)
SATTACH	−0.01648	0.00578	−2.850	0.00437	Each +1 unit: STRESS rate × 0.9837 (↓ 1.63%)

All four predictors are statistically significant at $\alpha = 0.05$, a noticeably stronger result than any previous model. Importantly, a standard Poisson regression assumes variance equals the mean ($Var[Y] = \mu$). We know from the initial distributional analysis that *STRESS* has variance ≈ 3.42 and mean ≈ 1.73, giving a variance-to-mean ratio of approximately 1.98, indicating overdispersion. If left unchecked, the Poisson model can underestimate standard errors and overstate significance.

4.2 Over-dispersed Regression

An over-dispersed or Negative Binomial model was fitted to predict *STRESS* scores with the same predictors as the standard Poisson model. A Negative Binomial model extends Poisson by adding a dispersion parameter θ (theta), which allows variance to exceed the mean. The table below reports the estimated, standard error, z-score, p-value, and IRR of each coefficient in the fitted model.

Term	Estimate	Std. Error	z-value	p-value	Interpretation (IRR)
(Intercept)	2.7590	0.3415	8.078	< 0.001	exp(2.7590) = 15.79; baseline rate when all X = 0
COHES	−0.01339	0.00414	−3.238	0.00121	Each +1 unit: STRESS rate × 0.9867 (↓ 1.33%)
ESTEEM	−0.02306	0.01148	−2.009	0.04453	Each +1 unit: STRESS rate × 0.9772 (↓ 2.28%)
GRADES	−0.02436	0.01397	−1.744	0.08118	Each +1 unit: STRESS rate × 0.9759 (↓ 2.41%)

SATTACH	−0.01675	0.00830	−2.019	0.04349	Each +1 unit: STRESS rate × 0.9834 (↓ 1.66%)

The coefficient point estimates are almost identical to the Poisson model — the log-linear relationship between predictors and expected stress is consistent across both specifications. The critical difference lies in the standard errors and the resulting inference. Moreover, the AIC of the Negative Binomial model was 133.6 units lower than Poisson's 2417.2, providing substantially better fit to the data.

The Negative Binomial regression corrects all major deficiencies identified in the OLS model. The log link and count distribution guarantee predicted stress scores are always positive, which the OLS model had negative fitted values. Additionally, heteroscedasticity is resolved by design of the count models as these models allow for variance to vary with the mean.

4.3 *STESS* with *COHES* Interpretation

Using the fitted Poisson regression model, predicted *STRESS* counts were computed for three levels of family cohesion (*COHES*): a low group (one standard deviation below the mean), a middle group (at the mean), and a high group (one standard deviation above the mean). All other predictors were held fixed at their sample means, isolating the effect of *COHES*. Predicted counts are obtained by computing the linear predictor (LP) for each group and applying the inverse log link (exponential function), which converts from the log scale to the expected count scale:

$$\mathbf{E[STRESS] = \exp(\beta_0 + \beta_1 \cdot \mathit{COHES} + \beta_2 \cdot \mathit{ESTEEM_mean} + \beta_3 \cdot \mathit{GRADES_mean} + \beta_4 \cdot \mathit{SATTACH_mean})}$$

This is the fundamental prediction equation of the Poisson model. Because all predictors other than *COHES* are held at their means, this is a marginal prediction. It represents the expected stress count for a hypothetical average respondent who differs only in their level of family cohesion. The table below reports the results of predicted *STRESS* counts for the family cohesion groups:

Group	*COHES* Value	Predicted *STRESS* Count	vs Low Group	vs High Group
Low	38.85	1.9148	—	+34.2% more than High
Middle	50.22	1.6530	-13.7% vs Low	-13.7% vs High
High	61.59	1.4269	-25.5% vs Low	—

Moving from the Low group to the Middle group reduces predicted stress by 13.7%, and moving from the Middle group to the High group reduces it by a further 13.7%. It follows directly from the log link function used in Poisson regression:

$$\mathbf{Predicted\ count = \exp(\beta_0 + \beta_1 \cdot \mathit{COHES} + ...) = \exp(constant) \times \exp(\beta_1 \cdot \mathit{COHES})}$$

Because the effect of *COHES* enters multiplicatively via $\exp(\beta_1)$, equal increments in *COHES* produce equal proportional (not equal absolute) changes in predicted stress. Each additional standard deviation of family cohesion multiplies expected stress by $\exp(-0.012918 \times 11.37) \approx$

0.863, a reduction of approximately 13.7%. The expected percentage difference between the High and Low groups is computed directly from the predicted counts:

$$\textbf{\% difference} = (1.4269 - 1.9148) / 1.9148 \times 100 = -25.5\%$$

Respondents in the High *COHES* group are expected to report 25.5% fewer stressful events than those in the Low *COHES* group. Equivalently, from the Low group's perspective, the High group experiences 34.2% more stress, the asymmetry arises because the denominator differs. A 25.5% reduction in stressful events between the low and high cohesion groups is a substantial effect for a single predictor, particularly given that the full model explains only ~8% of variance. Thus, *COHES* is the single strongest predictor of all four models tested.

4.4 Model Diagnostics

Both the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) decisively favor the Negative Binomial model over the standard Poisson model. The Negative Binomial achieves an AIC of 2283.590 versus 2417.219 for Poisson (ΔAIC = 133.629), and a BIC of 2310.461 versus 2439.612 (ΔBIC = 129.151). Differences of this magnitude, which far exceeding the conventional threshold of 10, constitute overwhelming evidence in favor of the Negative Binomial. Crucially, the BIC imposes a heavier penalty for model complexity than AIC, yet the Negative Binomial still wins by a wide margin despite having one additional parameter (the dispersion parameter θ).

4.5 Deviance Residuals vs Predicted Values

The deviance residuals vs. predicted values plot below for the Poisson regression model reveals a highly structured pattern of curved arcs, signaling the defining visual signature of a Poisson model applied to discrete count data. Each arc corresponds to a distinct observed value of *STRESS* (0, 1, 2, …, 9), and the shape, spacing, and density of the arcs communicate three substantive problems with the model: excess zeros, overdispersion, and systematic under-prediction at high stress values. The plot confirms that the standard Poisson model is incorrectly specified and reinforces the case for the Negative Binomial as the preferred model.

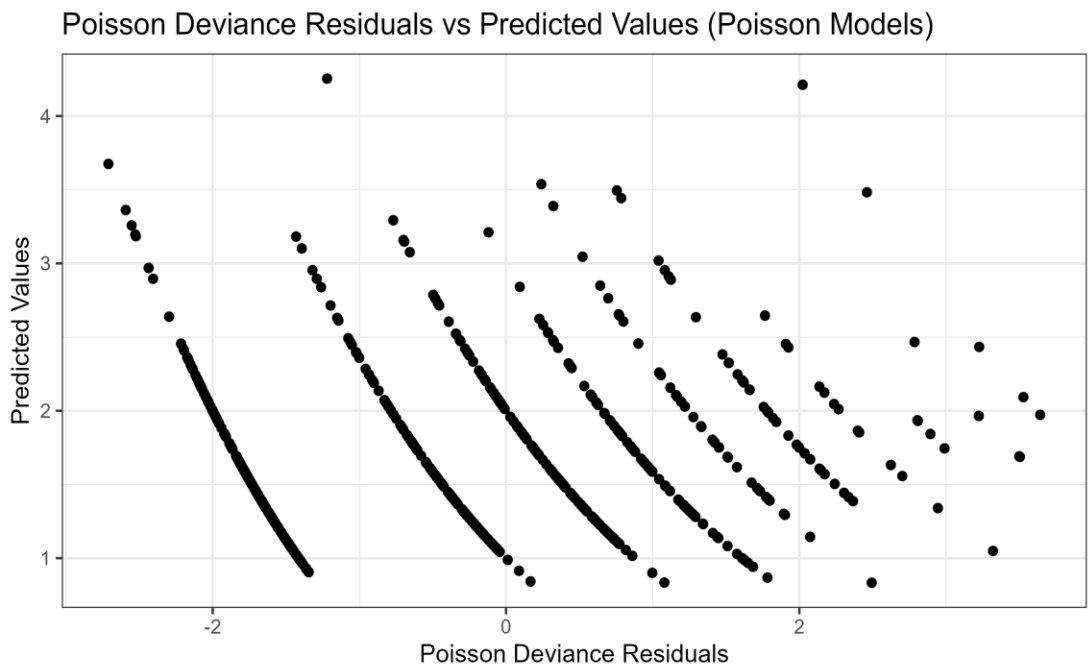

Figure 5. Poisson deviance residuals plotted against fitted values (n = 651). Each arc corresponds to one observed STRESS value.

While the arc pattern itself is expected, three aspects of this specific plot raise concerns beyond the normal appearance of a count model residual plot. First, the leftmost arc (y = 0) is unusually prominent and dense. Under a well-calibrated Poisson model with mean 1.73, only ~17.7% of observations would be expected at zero. The actual proportion is 34% (n = 221). This excess zero mass creates an abnormally heavy leftmost arc, indicating a direct visual indicator of the zero-inflation problem. Second, the arcs for y = 1 and y = 2 are also very dense (n = 128 and n = 125 respectively), clustering at predicted values of 1.0–2.5. Together with the zero arc, this creates an overloaded left and center region of the plot, reflecting the heavily right-skewed distribution of *STRESS*. Third, the right side of the plot (residuals > +2, predicted values > 2.5) contains sparse but widely scattered points corresponding to the highest *STRESS* scores (6–9). These points lie far from any arc, indicating the model has no reliable mechanism for predicting high-stress outcomes.

5.0 Logistic Regression Model

A logistic regression model was fitted to predict whether a respondent experiences any stress at all (*Y_IND* = 1 if *STRESS* > 0; Y_IND = 0 if *STRESS* = 0) using four predictors: COHES, *ESTEEM*, *GRADES*, and *SATTACH*. the estimated, standard error, z-score, p-value, and Odds Ratio (OR) of each coefficient in the fitted model.

Term	β	SE	z	p-value	OR	Interpretation
(Intercept)	3.5167	0.7371	4.771	< 0.001	33.674	Baseline log-odds when all X = 0

COHES	−0.0207	0.0088	−2.369	0.0178	0.9795	Each +1: odds × 0.9795 (↓ 2.1%)
ESTEEM	−0.0189	0.0237	−0.795	0.4268	0.9813	Each +1: odds × 0.9813 — NOT significant
GRADES	−0.0255	0.0287	−0.888	0.3744	0.9748	Each +1: odds × 0.9748 — NOT significant
SATTACH	−0.0277	0.0175	−1.582	0.1136	0.9727	Each +1: odds × 0.9727 — NOT significant

The model is statistically significant overall (LRT $\chi^2(4) = 22.39$, $p < 0.001$), but its practical explanatory power is very weak. When all predictors equal zero, the log-odds of any stress is 3.517, corresponding to a probability of exp(3.517)/(1+exp(3.517)) = 97.1%. This is not directly meaningful as all-zero predictor values are outside the observed range, but the large, significant intercept reflects the fact that stress is present in most of the sample.

Only *COHES* reaches conventional significance ($\beta = -0.0207$, OR = 0.9795, $p = 0.018$). Each one-unit increase in family cohesion reduces the odds of any stress by approximately 2.1%. *ESTEEM*, *GRADES*, and *SATTACH* are all non-significant ($p > 0.10$), suggesting these predictors are more relevant to predicting the intensity of stress (how many events) than to predicting its mere presence or absence. Therefore, the model should be rerun in a reduced form retaining only *COHES*, and multicollinearity among the predictors should be investigated before any further conclusions are drawn.

6.0 Manual ZIP Regression

The distribution of *STRESS* scores is best understood as the product of two overlapping generative processes: one determining whether a student is susceptible to stress at all, and a second determining how many stressful events occur among those who are susceptible. This structure motivates a Zero-Inflated Poisson (ZIP) model, which is constructed here manually by fitting a logistic regression to predict stress presence (*Y_IND*) and a Poisson regression to predict stress count conditional on stress being present (*STRESS_IND*). The combined predictions were used to determine the expected value of *STRESS*, the probability that a given student belongs to the at-risk (stress-susceptible) group (*P(stress present)*) weighted by the expected count conditional on stress being present (*E[STRESS | stress present]*).

6.1 Logistic Regression

The table below reports the model summary for the logistic regression to predict stress presence.

Term	β	SE	z	p	OR
(Intercept)	3.5167	0.7371	4.771	<0.001	33.74
COHES	−0.0207	0.0088	−2.369	0.018	0.979

ESTEEM	−0.0189	0.0237	−0.795	0.427	0.981
GRADES	−0.0255	0.0287	−0.888	0.374	0.975
SATTACH	−0.0277	0.0175	−1.582	0.114	0.973

The logistic component shows the same pattern as the standalone logistic model analyzed previously: only *COHES* is a significant predictor of whether stress is present at all. *ESTEEM*, *GRADES*, and *SATTACH* do not reliably discriminate between the stress-present and stress-absent groups. This is the weakest component of the ZIP model and is its primary limitation: if we cannot reliably identify who is in the at-risk group, the two-process structure adds limited value over a simpler single-component model.

6.2 Poisson Regression

The below reports the model summary of the Poisson regression to predict count conditional on stress being present.

Term	β	SE	z	p	IRR	Note
(Intercept)	2.3117	0.2400	9.633	<0.001	10.09	Baseline count when stress is present
COHES	−0.0063	0.0029	−2.124	0.034	0.994	↓ 0.6% events per unit — significant
ESTEEM	−0.0195	0.0081	−2.408	0.016	0.981	↓ 1.9% events per unit — significant
GRADES	−0.0147	0.0097	−1.513	0.130	0.985	Not significant (p=0.130)
SATTACH	−0.0080	0.0058	−1.371	0.170	0.992	Not significant (p=0.170)

The Poisson component was fitted on the 430 respondents with *STRESS* > 0, the 221 zero-stress respondents are excluded (coded as NA in *STRESS_IND*). This is conceptually correct: we are modelling the intensity of stress only among those for whom stress is possible. *COHES* primarily predict stress presence (logistic component significant). *ESTEEM* primarily predicts stress intensity given presence (Poisson component significant). This dissociation suggests the two predictors operate through different mechanisms.

6.3 Goodness-of-fit

For each respondent i, the ZIP predicted value is:

$$\hat{y}_i = \hat{P}(Y_IND_i = 1) \times \exp(\hat{\beta}_Poisson \cdot X_i)$$

where the logistic model provides $\hat{P}(Y_IND = 1)$ and the Poisson model provides $\hat{\lambda} = \exp(X\beta)$. Residuals are computed as $e_i = STRESS_i - \hat{y}_i$ on the original count scale. The model's predicted mean (1.730) is almost identical to the observed mean (1.730). This is mathematically expected by construction, $E[\hat{y}] = E[P(Y=1)] \times E[\lambda|Y=1] \approx E[Y]$ when the model is properly specified. It confirms the combination formula was applied correctly and that the model is unbiased on average.

The confusion matrix reveals the most serious problem with the manual ZIP approach:

	Actual: Stress Present (Y_IND=1)	Actual: No Stress (Y_IND=0)	Row Total
Predicted: Stress (prob ≥ 0.5)	422 – True Positive	210 – False Positive	632 (97.1% of all)
Predicted: No Stress (prob < 0.5)	8 – False Negative	11 – True Negative	19 (2.9% of all)
Column Total	**430 (66.0%)**	**221 (34.0%)**	**651 (100%)**

Only 19 respondents are predicted as zero-stress (predicted probability < 0.5), of whom 11 are correctly identified and 8 are false negatives. Of the 221 true zero-stress respondents, only 11 (5.0%) are correctly classified. This means 210 of them are incorrectly assigned to the stress-present group.

The RMSE of 1.768 is virtually identical to the OLS RMSE of 1.776, and the residual variance (3.130) is only 8.5% lower than the total variance of *STRESS* (3.419). The model reduces unexplained variance by less than one-tenth: a weak result for a two-component model that uses eight estimated parameters (five per equation, with intercept overlap).

7.0 *pscl* ZIP Regression Model

Four formal Zero-Inflated Poisson/Negative Binomial (ZIP/ZINB) models were fitted using the *pscl* package's *zeroinfl()* function. These models are superior to the manual ZIP approach because both components are estimated simultaneously via maximum likelihood, producing proper standard errors and valid information criteria. The first uses all four predictors on both sides of the model, providing a full baseline specification. The remaining three refine the model through variable selection, arriving at a preferred specification.

7.1 ZINB Full Model

The first formal ZIP model uses all four predictors (*COHES*, *ESTEEM*, *GRADES*, and *SATTACH*) in both the count and zero-inflation components, with a Poisson distribution for the count part. The model formula is:

STRESS ~ COHES + ESTEEM + GRADES + SATTACH | COHES + ESTEEM + GRADES + SATTACH

This full specification serves as a benchmark. Before any variable selection, it tells us which predictors contribute meaningfully to each process and where the model can be simplified. The model converged in a single BFGS iteration, with a log-likelihood of −1134 on 10 degrees of freedom, an AIC of 2288.8, and a BIC of 2333.6.

In the count component, two predictors are clearly significant: *COHES* ($\beta = -0.0083$, $z = -2.42$, $p = 0.016$) and *ESTEEM* ($\beta = -0.0261$, $z = -2.83$, $p = 0.005$). *GRADES* is marginally significant ($p = 0.073$), while *SATTACH* falls short of conventional thresholds ($p = 0.116$). Both significant predictors carry negative coefficients, confirming that higher family cohesion and higher self-esteem are associated with fewer stressful events among the at-risk group. In incidence rate ratio terms, each additional unit of *COHES* reduces the expected count by 0.8%, and each additional unit of *ESTEEM* reduces it by 2.6%.

The zero-inflation component is the most important result from this model, and it is sobering. Not one of the four predictors reaches statistical significance in predicting structural zero status. The intercept is significant ($\beta = -2.835$, $z = -2.88$, $p = 0.004$), confirming that a non-trivial proportion of the sample is structurally stress-free, but the available predictors cannot identify which students belong to this group. *COHES* has the most plausible coefficient direction (positive, meaning higher cohesion increases the probability of being a structural zero), which aligns with theory as highly cohesive families may be genuinely protective enough to keep some students entirely outside the stress-susceptible pool, but the effect is not statistically distinguishable from zero ($p = 0.119$).

7.2 Best Fitting ZIP Model: ZINB Reduction Model

Based on the results of the baseline model, *GRADES* and *SATTACH* were removed from both components as non-significant. *COHES* was retained on the zero-inflation side as the predictor closest to significance in that component. The count distribution was upgraded from Poisson to Negative Binomial, motivated by the persistent overdispersion observed across all previous models. The resulting specification is:

***STRESS* ~ *COHES* + *ESTEEM* | *COHES*, dist = 'negbin'**

This model uses six parameters, five fewer than the baseline model, and achieves an AIC of 2273.7 and BIC of 2300.6, both substantially better than the full model. The log-likelihood is −1131 on 6 degrees of freedom.

With the Negative Binomial distribution, both *COHES* and *ESTEEM* are clearly significant in the count component. *COHES* carries a coefficient of −0.0119 ($z = -2.98$, $p = 0.003$), corresponding to a 1.18% reduction in expected stress per unit increase. *ESTEEM* is the stronger predictor, with a coefficient of −0.0336 ($z = -3.29$, $p < 0.001$), implying a 3.3% reduction per unit. Self-esteem therefore appears to be particularly important in shaping the intensity of stress once it occurs, whereas its role in determining whether stress is present at all is less clear. The dispersion parameter theta is estimated as 5.374 ($\log(\theta) = 1.682$, $z = 4.96$, $p < 0.001$), which is substantially larger than the theta of 1.865 obtained in the standalone Negative Binomial model. This increase occurs because the zero-inflation component is now accounting for a portion of the excess zeros, leaving the count component to model a less overdispersed distribution.

COHES in the zero-inflation component carries a positive coefficient of +0.0238 ($z = 1.70$, $p = 0.089$), placing it at the margin of conventional significance. The theoretical interpretation is coherent: higher family cohesion increases the probability that a student is structurally stress-free. While the evidence is not conclusive, the marginal result is consistent across multiple model specifications and warrants retention of this predictor in the zero-inflation component for theoretical reasons.

7.3 Best Fitting ZIP Model: ZINB with Intercept-Only Zero Inflation

Because the zero-inflation predictors consistently fall short of significance, a natural question is whether the zero-inflation structure is better captured by a constant proportion, a single

intercept, rather than by predictor-driven probabilities. This specification acknowledges that structural zeros exist in the data while conceding that the model cannot identify which individuals belong to the zero-inflation group. It is a statistically honest representation of what the data supports.

The first model uses *COHES* and *ESTEEM* on the count side with an intercept-only zero-inflation component. It is the most parsimonious of the four formal ZIP models, using only five parameters. The AIC of 2274.6 is marginally higher than the reduced model, but the BIC of 2297.0 is the lowest of all four models, reflecting the benefit of a simpler zero-inflation specification. The zero-inflation intercept is −1.513 ($p < 0.001$), implying:

$$\mathbf{P(structural\ zero) = \exp(-1.513) / (1 + \exp(-1.513)) \approx 0.181}$$

The model therefore estimates that approximately 18% of students are structurally stress immune. This is a substantively important estimate. It lies between the naive zero count in the data (34%) and the Poisson-predicted zero proportion (18%), and it suggests that roughly half the observed zeros are structural while the other half are sampling zeros, suggesting that susceptible students who happened not to experience any stressful events during the measurement period.

The second ZINB model added back school attachment (*SATTACH*) to the count side to test whether its earlier non-significance was an artefact of model misspecification rather than a genuine absence of effect. In this model, *SATTACH* is statistically significant ($\beta = -0.0166$, $z = -2.25$, $p = 0.025$), as are *COHES* ($p = 0.001$) and *ESTEEM* ($p = 0.006$). The zero-inflation intercept remains the same in spirit, and the model achieves an AIC of 2271.5, the lowest of all four formal ZIP models and the lowest AIC of every model tested across the entire analysis. The BIC of 2298.4 is slightly higher than the previous ZINB model, reflecting the one extra parameter.

The table below presents the AIC and BIC for all four formal ZIP/ZINB models. Both criteria agree that the full ZIP with all four predictors on both sides (full model) is the worst-performing specification despite having the most parameters, clear evidence of overparameterization in the zero-inflation component.

Model Specification	df	AIC	BIC	AIC Rank	BIC Rank
ZIP — all 4 predictors on both sides	10	2288.8	2333.6	3rd	4th
ZINB — COHES + ESTEEM | COHES	6	2273.7	2300.6	2nd	3rd
ZINB — COHES + ESTEEM | constant	**5**	**2274.6**	**2297.0**	**3rd**	**1st**
ZINB — COHES + ESTEEM + SATTACH | constant	**6**	**2271.5**	**2298.4**	**1st**	**2nd**

AIC selects the final model (ZINB with *COHES*, *ESTEEM*, *SATTACH* on the count side and intercept-only zero inflation) as the preferred specification. BIC selects first ZINB ZIP model (the same structure without *SATTACH*) as the most parsimonious adequate fit. The disagreement between the two criteria is narrow, just 1.4 AIC units separate the two, and hinges solely on whether *SATTACH* is included. Given that *SATTACH* is statistically significant in the final model

and theoretically motivated by the school attachment literature, the final model is the preferred specification for inference. The first ZINB ZIP model is the preferred specification when parsimony is the overriding concern.

7.4 Conclusions

Nine models have now been fitted to the *STRESS* variable, progressing from simple OLS through log-transformation, Poisson, Negative Binomial, a manual ZIP, and finally a family of formally estimated ZINB models. Each step has added a piece of the picture. Taken together, they tell a coherent and consistent story.

The first two models, OLS and log-transformed OLS, established a clear negative result. *STRESS* is a discrete, bounded, right-skewed, zero-inflated count variable, and neither OLS nor its log-transformed variant can accommodate these features. Heteroscedasticity was severe and structural in the OLS model, persisting in a different form after transformation. The log-transformed model also introduced back-transformation bias and saw *ESTEEM* lose its statistical significance. Both should be regarded as instructive failures that motivated the move to count regression.

The Poisson model was the first correctly specified model for the outcome type. It was revealed that all four predictors were negatively associated with stress and that the overall model was highly significant. However, the residual deviance-to-degrees-of-freedom ratio of 1.93 confirmed substantial overdispersion, meaning the Poisson underestimated standard errors by approximately 39 percent and overstated the significance of every predictor. The Negative Binomial model corrected this by estimating a dispersion parameter $\theta = 1.865$, which closely matched the observed variance-to-mean ratio. *GRADES* dropped below conventional significance thresholds once overdispersion was properly handled, a meaningful correction. With an AIC of 2283.6, Negative Binomial is the best single equation count model for this data.

The manual ZIP exercise was pedagogically important even though its practical performance was limited. By fitting the logistic and Poisson components independently, it revealed a key dissociation that single-equation models obscure: *COHES* predicts whether stress is present at all, while *ESTEEM* predicts how intense that stress is once it occurs. These two processes operate through distinct mechanisms and may benefit from distinct interventions. The manual ZIP also confirmed that the model correctly calibrated the mean (predicted mean = 1.730, actual = 1.730) but failed to identify zero-stress respondents reliably, classifying only 5 percent of them correctly, because the logistic component lacked sufficient discriminative power.

The formal ZINB models fitted via *pscl::zeroinfl()* resolved several ambiguities that the earlier models could not. Most importantly, they confirmed that while structural zeros are present in the data, the zero-inflation intercept is consistently and highly significant across all specifications, the available predictors cannot reliably identify which students belong to the structurally stress-free group. No predictor reaches conventional significance in the zero-inflation component once all four are included. *COHES* is marginally significant ($p = 0.089$) in a reduced specification, but this evidence is not strong enough to draw firm conclusions. The best

representation of the zero-inflation structure is therefore a constant proportion, estimated at approximately 18 percent of the sample.

Within the count component of the ZINB models, *COHES* and *ESTEEM* emerge as the most robust predictors. Both are consistently significant across all ZINB specifications, and their effects are somewhat larger than in the unconditional models because the zero-inflation component is now absorbing the structural zeros that previously diluted the count coefficients. *SATTACH* re-enters significance in final model, the most complex of the formal ZINB models, with a p-value of 0.025. Its marginal status in earlier models is now interpretable because of zero-inflation bias rather than a genuine absence of effect. *GRADES*, by contrast, never achieves significance in any count component of the ZINB family and is consistently the weakest predictor across all model specifications. It is the only predictor for which there is genuinely weak evidence.

The recommended model for inference and reporting is the first ZINB ZIP model: a Zero-Inflated Negative Binomial regression with *COHES* and *ESTEEM* in the count component and an intercept-only zero-inflation term. Its specification is:

zeroinfl(*STRESS* ~ *COHES* + *ESTEEM* | 1, dist = 'negbin')

This model achieves the lowest BIC of all nine models tested (2297). Both count predictors are statistically significant, the Negative Binomial distribution correctly handles overdispersion, and the zero-inflation structure is modelled honestly given the available evidence. Furthermore, *COHES* and *ESTEEM* have been proved as robust predictors across the nine models.